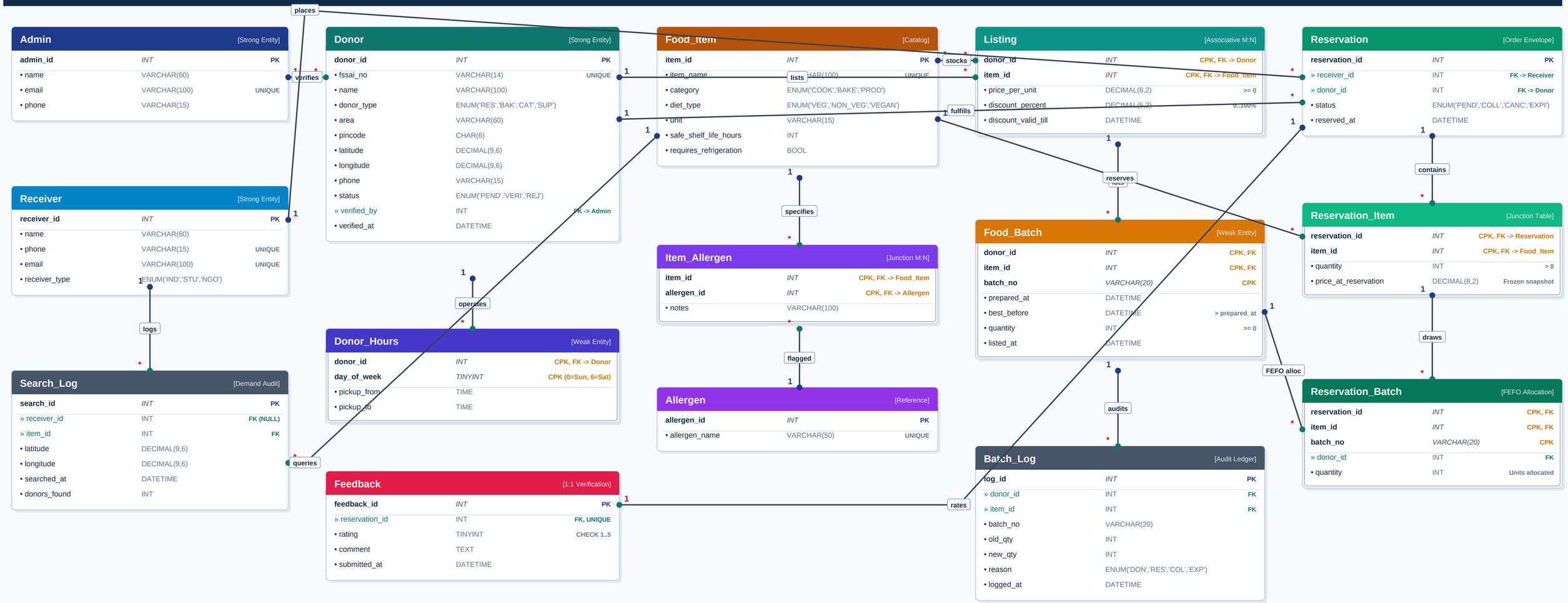
SURPLUS FOOD RESCUE — RELATIONAL ENTITY-RELATIONSHIP SCHEMA DIAGRAM

CECS202 DBMS Mini Project · Sardar Patel Institute of Technology (SPIT) · Batch B2



ARCHITECTURAL SCHEMA CONVENTIONS, INTEGRITY CONSTRAINTS & 3NF NORMALIZATION

Key Notations & Integrity

- PK: Primary Key — Unique identifier (NOT NULL)
- CPK: Composite Primary Key in weak/junction tables
- FK: Foreign Key referencing parent relation (Referential)
- UNIQUE: Candidate key preventing duplicate business facts
- CHECK: Domain constraints (e.g. qty >= 0, rating 1..5)

Cardinality Semantics

- 1 : N (One-to-Many): Single parent entity row references multiple child records (e.g. 1 Donor -> many Listings)
- M : N (Many-to-Many): Resolved via associative tables (Listing, Item_Allergen, Reservation_Item)
- 1 : 1 (One-to-One): Strict single review per Reservation

Entity Classifications

- Strong Entities (Single border): Independent existence
- Weak Entities (Double border): Existence dependent on identifying owner (Donor_Hours -> Donor, Food_Batch)
- Associative / Junction Relations: Maintain transaction snapshots (e.g. price_at_reservation frozen in time)

Relational Normalization

- 1NF: Atomic attributes, zero repeating groups
- 2NF: No partial dependency on composite keys
- 3NF: Zero transitive dependencies on non-key columns
- Target Engine: MySQL 8.0+ / 9.4.0 InnoDB ACID
- Total Tables: 15 Relations | 0 Redundant Attributes